

Yaohui Shu

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SUMMARY

Ph.D. candidate in Applied Mathematics and Statistics studying how correlated noise, memory, and constraints shape outcome distributions in large stochastic populations, and how to tell cause from correlation in such systems. Builds minimal SDE models, derives exact results, validates them with large-scale Monte Carlo, and uses solvable cases as ground truth for causal inference. Applied to turbulent cloud physics (first-author paper, *Geophysical Research Letters*).

EDUCATION

Stony Brook University, Ph.D., Applied Mathematics & Statistics (GPA 3.78 / 4.0) Expected May 2027
Dissertation: *Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods*
Coursework: Advanced Stochastic Models & Risk Analysis, Numerical Analysis I-III, Financial Derivatives & Stochastic Calculus
University of Rochester, M.S., Finance (GPA 3.81 / 4.0) Aug 2021 – Aug 2022
Coursework: Derivatives Pricing, Financial Econometrics, Risk Management, Stochastic Calculus for Finance
CQF Institute, Certificate in Quantitative Finance (CQF)

SKILLS

Programming	Python (NumPy, SciPy, pandas, TensorFlow), MATLAB, C++, R; Git, SLURM/HPC
Simulation	Explicit and implicit SDE integrators, Monte Carlo with time-step and sample-size convergence testing, constrained boundary handling, large parameter sweeps on HPC
Stochastic Modeling	Stochastic calculus, Fokker-Planck and Langevin equations, Ornstein-Uhlenbeck processes, maximum entropy, Weibull and gamma distribution families
Statistics & ML	Causal discovery (FCI, PCMC, NOTEARS), kNN conditional mutual information, transfer entropy, doubly robust estimation (DML, AIPW, TMLE), LSTM, XGBoost, PCA
Finance	Derivatives pricing, financial econometrics, risk management, equity signal research, backtesting

RESEARCH EXPERIENCE

Stony Brook University Stony Brook, NY
Doctoral Researcher: Stochastic Modeling, Monte Carlo & Causal Inference Aug 2023 – Present

- Lead, as the group's sole Ph.D. student, a research program on stochastic populations driven by a shared fluctuating environment under conservation laws, covering theory, simulation, causal analysis, and publication.
- Reconciled maximum-entropy statistics and Langevin dynamics for the same system: traced their disagreement to the choice of state variable and generalized the SDE model to any conserved moment order under one hypothesis.
- Identified a structural trade-off in finite-memory stochastic models (no boundary rule can conserve probability, preserve the driving process, and avoid mass at zero at once) and traced its effects in Monte Carlo results.
- Designing a causal discovery framework for nonstationary time series with feedback loops and unobserved drivers (partial ancestral graphs, nearest-neighbor independence tests), with a path-space entropy rate as a causal-strength measure.
- Built stochastic benchmarks for how mixing with an external environment reshapes a population, spanning the continuum between two limiting regimes.

Teaching Assistant, AMS 595 Fundamentals of Computing (MATLAB, Python, C++) Aug 2026 – Present

Brookhaven National Laboratory Upton, NY
Research Assistant: Stochastic Processes, Monte Carlo & Information Theory Jan 2024 – Jan 2026

- Showed where the white-noise diffusion approximation fails: extended it to SDEs with finite-memory (Ornstein-Uhlenbeck) noise, recovering the classical limit and quantifying how memory and intensity drive dispersion.
- Turned a solvable stochastic model into an exact benchmark for causal inference: derived time-resolved information flow in closed form and built an ensemble estimator that recovers causal direction where correlation cannot.
- Found that dispersion collapses onto the ratio of noise intensity to decorrelation rate, and that causal coupling between driver and population fades as the distribution spreads; published in *Geophysical Research Letters*.

CQF Institute Remote
Quantitative Finance Capstone: Equity Return Prediction 2023

- Built an LSTM direction model from 156+ momentum, volatility, and technical features, cut to 10–15 via XGBoost importance, self-organizing maps, and PCA; leakage-free splits and a 162-configuration search.

PUBLICATIONS & MANUSCRIPTS

- **Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., Li, X. (2026). Stochastic Condensation with Information Flow for Causality Analysis. *Geophysical Research Letters*, 53(17), e2026GL123056. doi:10.1029/2026GL123056
- **Shu, Y.**, Liu, Y., Li, X. A Langevin-Type Stochastic Condensation Model to Explain the Size Distribution based on Maximum Entropy Principle. In preparation for *Physical Review E*.
- **Shu, Y.** Entropy-Based Causal Discovery for Spatiotemporal Dynamical Systems with Feedback and Unmeasured Drivers. Methodology working paper; one further manuscript in preparation.
- Conference abstracts: AGU Fall Meeting 2025 (A21B-2034, first author; A21B-203); 106th AMS Annual Meeting 2026.